Job Sequencing with Greedy Method

Algorithm -: ① Arrange All in descending order of Profit ($n \log n$)

② For each job (m_i), do linear search to find particular slot in array of size (n), where n = maximum deadline

m = total jobs

Jobs	J_1	J_2	J_3	J_4
Profit	50	15	10	25
Deadline	2	1	2	1

Uniprocessor. No preemption

1 unit of time (every time)

↓

2 वाले पहले भी (slot 1) मे काम कर सकता है और 2 के में तो फिर ही रखता है

	J_1	J_4	J_2	J_3
① Arrange desc	50	25	15	10
	2	1	1	2

② ज्यादा deadline वाले slot को दूर से दूर वाला slot मान डालो. eg - 50(2) वाले को 2nd slot me डालो

25	50
0	1 2

50(2nd) slot मे डालने के बाद next बड़ा element है 25(1) But it can be added in 1st slot अगर 2nd होता तो नहीं कर सकते थे।

So total profit => 50 + 25

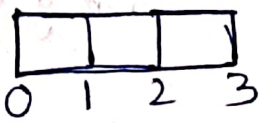
= 75

$n = 5$

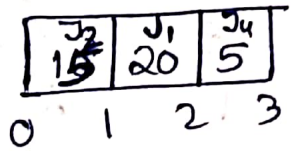
Jobs	J_1	J_2	J_3	J_4	J_5
Profit	20	15	10	5	1
deadline	2	2	1	3	3

Step 1) Arrange in descending order (already arranged) according to Profit

Step 2) make box till max unit of deadline



$\Rightarrow$ Put values as discussed in previous page



Total Profit = 15 + 20 + 5
= 40 unit

Step 3 Make Table

Job Considered	Slot	Solution	Profit
—	—	—	0
J_1 ✓	[1,2]	J_1	20
J_2 ✓	[0,1] [1,2]	J_2, J_1	20+15
J_3 ✗	[0,1] [1,2] No Empty Slot for 0 to 1	J_2, J_1	20+15
J_4 ✓	[0 1] [1 2] [2,3] Last slot [2,3]	J_2, J_1, J_3	20+15+5
J_5	[0 1] [1 2] [2 3] No Empty slot	J_2, J_1, J_3	20+15+5

$\boxed{\text{Total Profit} = 40}$

Example-2 Job Sequencing with Deadline

$n = 7$

Job	J_1	J_2	J_3	J_4	J_5	J_6	J_7
Profit	35	30	25	20	15	12	5
deadline	3	4	4	2	3	1	2

Step 1) Arrange in descing order of Profit (already arranged)

Step 2) Max deadline = 4

J_4	J_3	J_1	J_2
20	25	35	30

0 1 2 3 4

Total Profit = 110

Step 3)

Job Considered	Slot	Solution	Profit
J_1	(2,3)	J_1	35 = 35
J_2	(2,3)(3,4)	J_1, J_2	35+30 = 65
J_3	(12)(23)(34)	$J_3\ J_1\ J_2$	35+30+25 = 90
J_4	(01)(12)(23)(34)	$J_4\ J_3\ J_1\ J_2$	35+30+25+20 = 110
J_5	Slot full	$J_4\ J_3\ J_1\ J_2$	110
J_6	Slot full	$J_4\ J_3\ J_1\ J_2$	110
J_7	Slot full	$J_4\ J_3\ J_1\ J_2$	110

Total Profit = 110